APÉRY POLYNOMIALS AS STIELTJES MOMENT SEQUENCES

KNOWLEDGE SENSE

ABSTRACT. For real x let $A_n(x) = \sum_{j=0}^{n} \binom{n}{j}^2 \binom{n+j}{j}^2 x^j$ be the Apéry polynomials. Sokal conjectured that $(A_n(x))_{n\geq 0}$ is a Stieltjes moment sequence for every $x \geq 1$; the case $x = 1$, concerning the Apéry numbers, was proved by Edgar. We prove the conjecture. For every real $x \geq 1$ there is a probability measure μ_x on $[0, c_x]$, where $c_x = \bigl(1 + 2\sqrt{x} + 2\sqrt{x+\sqrt{x}}\bigr)^2$, such that $A_n(x) = \int t^n \, \mathrm{d}\mu_x(t)$ for all $n \geq 0$. The measure is absolutely continuous, and its density is given explicitly by one-dimensional integrals of complete elliptic integrals; it is strictly positive on $(0, c_x)$ and has at most a squared logarithmic singularity at the origin. The proof writes the generating function of $x^{-n}A_n(x)$ as an average of a resolvent over a three-dimensional torus, continues the resulting Stieltjes transform through a Joukowski-type change of variables, and identifies the boundary values on the second sheet by means of the logarithmic monodromy of the complete elliptic integral.

1. INTRODUCTION

A sequence $(a_n)_{n\geq 0}$ of real numbers is a *Stieltjes moment sequence* if there is a positive Borel measure μ on $[0, \infty)$ such that $a_n = \int t^n \, \mathrm{d}\mu(t)$ for all $n \geq 0$; equivalently, the Hankel matrix $(a_{i+j})_{i,j\geq 0}$ is totally positive [10]. Many sequences arising in enumerative combinatorics are Stieltjes moment sequences; see [7, 10] and the references there.

The Apéry numbers $A_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2$ appear in Apéry's proof of the irrationality of $\zeta(3)$ [1]. Sokal conjectured that they form a Stieltjes moment sequence, and Edgar [5] proved this, with a representing density expressed through Heun functions. The *Apéry polynomials*

$$A_n(x) = \sum_{j=0}^{n} \binom{n}{j}^2 \binom{n+j}{j}^2 x^j \qquad (n \geq 0) \tag{1.1}$$

satisfy $A_n(1) = A_n$. Sokal [7, Introduction, (1.2)] recorded as open the more general conjecture that $(A_n(x))_{n\geq 0}$ is a Stieltjes moment sequence for every

Date: 13 October 2026.

2020 *Mathematics Subject Classification.* Primary 44A60; Secondary 05A10, 11B65, 30E05, 33E05.

Key words and phrases. Apéry polynomials, Stieltjes moment sequence, Stieltjes transform, analytic continuation, complete elliptic integral.

$x \geq 1$, and the conjecture is restated as Conjecture 1 in [8, slide 29]. Our main result settles it.

Theorem 1.1. *Let $x \geq 1$ be real and put*

$$c_x = \Bigl(1 + 2\sqrt{x} + 2\sqrt{x + \sqrt{x}}\Bigr)^2. \tag{1.2}$$

There is a unique positive Borel measure μ_x on $\mathbb{R}$ such that

$$A_n(x) = \int t^n \, \mathrm{d}\mu_x(t) \qquad \text{for all } n \geq 0.$$

It is a probability measure, it is absolutely continuous with respect to Lebesgue measure, and its support is $[0, c_x]$. Its density φ_x, given explicitly in (1.9) below, is strictly positive on $(0, c_x)$ and satisfies $\varphi_x(t) = O(\log^2 t)$ as $t \to 0^+$. In particular, $(A_n(x))_{n \geq 0}$ is a Stieltjes moment sequence.

The density. We use the complete elliptic integral of the first kind in the *parameter* convention,

$$\mathbf{K}(m) = \int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{1 - m \sin^2 \theta}} \qquad (m < 1), \tag{1.3}$$

so that $\mathbf{K}(m)$ is $K(\sqrt{m})$ in the modulus notation of [4]. Let $x \geq 1$ and put $b = 2x^{-1/2} \in (0, 2]$ and $B = 4 + b$. Define

$$f(c) = \frac{1}{2\pi^2} \mathbf{K}\Bigl(1 - \frac{c^2}{16}\Bigr) \quad (0 < |c| < 4), \qquad f(c) = 0 \quad (|c| > 4), \tag{1.4}$$

and, for real w,

$$h(w) = \frac{1}{\pi} \int_{-b}^{b} \frac{f(w - u)}{\sqrt{b^2 - u^2}} \, \mathrm{d}u. \tag{1.5}$$

For real $w > b$ define further

$$T(w) = \frac{1}{2\pi^3} \int_{-b}^{b} \mathbf{K}\Bigl(1 - \frac{(w-u)^2}{16}\Bigr) \frac{\mathrm{d}u}{\sqrt{b^2 - u^2}}, \tag{1.6}$$

$$J(w) = \frac{1}{\pi^3} \int_{b}^{w} \kappa\Bigl(\frac{(w-u)^2}{16}\Bigr) \frac{\mathrm{d}u}{\sqrt{u^2 - b^2}}, \tag{1.7}$$

where

$$\kappa(m) = \mathbf{K}(m) \quad (0 \leq m < 1), \qquad \kappa(m) = m^{-1/2} \mathbf{K}(1/m) \quad (m > 1). \tag{1.8}$$

Finally let $r_x(t) = x^{-1/2}\bigl(\sqrt{t} + 1/\sqrt{t}\bigr)$ and

$$\varphi_x(t) = \begin{cases} \dfrac{h(r_x(t))}{\sqrt{xt}}, & 1 \leq t < c_x, \\[2ex] \dfrac{2T(r_x(t)) - h(r_x(t)) + 2J(r_x(t))}{\sqrt{xt}}, & 0 < t < 1, \end{cases} \tag{1.9}$$

and $\varphi_x(t) = 0$ for $t \notin (0, c_x)$. The values of f at the finitely many excluded points are irrelevant. All integrals converge absolutely: the only singularities are inverse square roots at the endpoints and logarithmic singularities of $\mathbf{K}$

at the parameter value 1. The function r_x decreases from $+\infty$ to b on $(0,1]$ and increases from b to B on $[1,c_x]$; at $t=1$ both lines of (1.9) equal $h(b)/\sqrt{x}$.

The positivity of φ_x is elementary. Since $f>0$ on $(-4,4)$, we have $h>0$ on $(-B,B)$. For $w>b$ the integrand of $T(w)$ is positive on all of $(-b,b)$, whereas $h(w)$ retains it only where $w-u<4$; thus $0\le h\le T$, and $\kappa>0$ gives $J\ge 0$. Hence $\varphi_x(t)\ge T(r_x(t))/\sqrt{xt}>0$ for $0<t<1$. What has to be proved is that the measure $\varphi_x(t)\,\mathrm{d}t$ has the moments $A_n(x)$, with exactly the combination $2T-h+2J$ on $(0,1)$.

Relation to earlier work. For $x=1$ Theorem 1.1 is Edgar's theorem [5, Theorem 1], and $c_1=17+12\sqrt{2}$ is the endpoint found there [5, Notation 3]; by uniqueness, μ_1 is Edgar's measure. The present paper extends the existence of a representing measure to every $x>1$ and gives the explicit density (1.9); the argument treats all $x\ge 1$ in the same way.

The proof starts from the polynomials $D_n(\alpha)=\alpha^n A_n(1/\alpha)$. By Straub's theorem on multivariate Apéry numbers [9, Theorem 3.1], applied with the partition $\lambda=(2,2)$, $D_n(\alpha)$ is the diagonal coefficient of $u^nv^nw^nz^n$ in

$$\frac{1}{(1-u-v)(1-w-z)-\alpha uvwz}.$$

An average of this rational function over a torus leads to the spectral representation of Proposition 3.1 (Remark 3.2).

Neuschel [6, Theorem 4.1] determined the asymptotic behaviour of the Apéry polynomials in $\mathbb{C}\setminus(-\infty,0]$; in particular $A_n(x)^{1/n}\to c_x$ for $x>0$. For a positive measure with compact support the nth roots of the moments converge to the largest point of the support, so once a representing measure is known to exist, the right endpoint c_x of its support is already determined by [6].

Two related questions are not addressed here. Sokal also conjectures that $(A_n(x))_{n\ge0}$ is not a Stieltjes moment sequence for $0<x<1$ [8, slide 29]; in our proof the hypothesis $x\ge1$ is essential for the holomorphic continuation of the spectral density across the central interval (Remark 4.3). His Conjecture 2 in [8, slide 29] asserts that $(A_n(1+y))_{n\ge0}$ is coefficientwise Hankel-totally positive in y, that is, that the minors of its Hankel matrix are polynomials in y with nonnegative coefficients. This would imply that $(A_n(x))_{n\ge0}$ is a Stieltjes moment sequence for every $x\ge1$, but it is a different, coefficientwise statement, and it remains open.

Strategy. Write $\alpha=1/x\in(0,1]$, so that $b=2\sqrt{\alpha}$, and let $D_n(\alpha)=\alpha^nA_n(1/\alpha)$.

(i) *A spectral representation.* Let $L=4\cos\theta_1\cos\theta_2+b\cos\theta_3$ with independent uniform angles. Then

$$\textstyle\sum_n D_n(\alpha)\tau^{2n}=\mathbb{E}\big[(1-\tau L+\alpha\tau^2)^{-1}\big]$$

(Proposition 3.1), and L has the density h.

(ii) *A folded Stieltjes transform.* With $s = \tau^{-2}$ this becomes

$$\textstyle\sum_n D_n(\alpha) s^{-n-1} = g(r)/\sqrt{s},$$

where g is the Cauchy transform of h and $r = \sqrt{s} + \alpha/\sqrt{s}$. The map $s \mapsto r$ is two-to-one: it sends the upper half-disc $\{|s| < \alpha,\ \operatorname{Im} s > 0\}$ into the lower half-plane and the semicircle $|s| = \alpha$ to the interval $(0, b)$. Continuing the transform into the half-disc therefore requires continuing g across $(0, b)$, that is, a holomorphic continuation of h from $(-b, b)$.

(iii) *The continuation.* In Section 4 we construct this continuation H explicitly in the lower half-plane. On the central interval $(-b, b)$ the condition $b \le 2$ keeps all differences of points of $(-b, b)$ inside $(-4, 4)$, and the jump of $\mathbf{K}$ across its branch cut, with its exact sign, shows that H is real there. To the right of b the boundary values of H have real part $T + J$.

(iv) *Reconstruction.* In Section 5 the Stieltjes transform is shown to be holomorphic on $\mathbb{C} \setminus [0, S]$, with growth bounds at the origin that exclude atoms, and Cauchy's integral formula recovers the measure from the jump across the cut.

Dependencies and organization. Besides elementary complex analysis, the proof uses only classical facts on the complete elliptic integral and the hypergeometric equation, which we collect in Section 2 with references to [4]. The results of [5, 6, 9] and of [2, 3] recalled in this paper are not used in the proofs: Proposition 3.1 is verified directly from the coefficients, and the endpoint c_x is obtained from the construction. Section 3 proves the spectral representation and the regularity of h. Section 4 constructs the continuation H and proves the bounds needed later. Section 5 studies the Stieltjes transform and proves Theorem 1.1. Section 6 contains further remarks.

2. The complete elliptic integral

For $m \in \mathbb{C} \setminus [1, \infty)$ and $\theta \in [0, \pi/2]$ the number $1 - m \sin^2\theta$ does not lie in $(-\infty, 0]$. With the principal square root, (1.3) therefore defines a holomorphic function on $\mathbb{C} \setminus [1, \infty)$, which is positive on $(-\infty, 1)$. Expanding the integrand gives $\mathbf{K}(m) = \frac{\pi}{2} \sum_{j \ge 0} \big(\frac{(1/2)_j}{j!}\big)^2 m^j$ for $|m| < 1$, so $\frac{2}{\pi}\mathbf{K}$ is the Gauss function ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; m)$ and $\mathbf{K}$ satisfies the hypergeometric equation

$$m(1-m)y'' + (1-2m)y' - \tfrac{1}{4}y = 0 \tag{2.1}$$

[4, Section 15.10]. Its singular points 0, 1, ∞ are regular, with exponents $\{0, 0\}$, $\{0, 0\}$ and $\{\frac{1}{2}, \frac{1}{2}\}$ respectively. In particular $\mathbf{K}$ continues analytically across $(1, \infty)$ from either side, so the boundary values $\mathbf{K}(m \pm i0) = \lim_{\varepsilon \downarrow 0} \mathbf{K}(m \pm i\varepsilon)$ exist and are continuous for $m > 1$.

Lemma 2.1 (Logarithmic expansion)**.** *There is a power series* $E(\varepsilon) = \sum_{j\ge 0} e_j \varepsilon^j$ *with real coefficients, convergent for* $|\varepsilon| < 1$*, such that*

$$\mathbf{K}(m) = \frac{1}{\pi}\,\mathbf{K}(1-m)\log\frac{1}{1-m} + E(1-m) \tag{2.2}$$

for $|1-m| < 1$*,* $m \notin [1,\infty)$*, with the principal logarithm.*

Proof. In the modulus notation, [4, 19.12.1] states that

$$K(k) = \sum_{j\ge 0}\Big(\frac{(1/2)_j}{j!}\Big)^2 k'^{2j}\Big(\log\frac{1}{k'} + d(j)\Big)$$

for $0 < k < 1$, with $k' = \sqrt{1-k^2}$ and explicit real numbers $d(j)$, the series converging for $|k'| < 1$. Put $m = k^2$, so that $k'^2 = 1-m$ and $\log(1/k') = \frac{1}{2}\log\frac{1}{1-m}$. Since $\sum_j \big(\frac{(1/2)_j}{j!}\big)^2 (1-m)^j = \frac{2}{\pi}\mathbf{K}(1-m)$, this is (2.2) for $0 < m < 1$, with $e_j = \big(\frac{(1/2)_j}{j!}\big)^2 d(j)$. Both sides are holomorphic on $\{|1-m|<1\}\setminus[1,2)$, so the identity persists there. □

Lemma 2.2 (Real part on the cut)**.** *For* $m > 1$*,*

$$\operatorname{Re}\mathbf{K}(m+i0) = \operatorname{Re}\mathbf{K}(m-i0) = m^{-1/2}\,\mathbf{K}(1/m) > 0.$$

Proof. Let $\theta_0 = \arcsin(m^{-1/2})$. As $\varepsilon \downarrow 0$, the integrand $(1-(m\pm i\varepsilon)\sin^2\theta)^{-1/2}$ converges to the positive number $(1-m\sin^2\theta)^{-1/2}$ for $\theta < \theta_0$, and to a purely imaginary number for $\theta > \theta_0$. Since $|1-(m\pm i\varepsilon)\sin^2\theta| \ge |1-m\sin^2\theta|$, the integrands are dominated by the integrable function $|1-m\sin^2\theta|^{-1/2}$. Hence, with the substitution $\sin\theta = m^{-1/2}\sin\phi$,

$$\operatorname{Re}\mathbf{K}(m\pm i0) = \int_0^{\theta_0}\frac{\mathrm{d}\theta}{\sqrt{1-m\sin^2\theta}} = \frac{1}{\sqrt{m}}\int_0^{\pi/2}\frac{\mathrm{d}\phi}{\sqrt{1-m^{-1}\sin^2\phi}}.$$ □

The full boundary values are $\mathbf{K}(m\pm i0) = m^{-1/2}\big(\mathbf{K}(1/m) \pm i\,\mathbf{K}(1-1/m)\big)$, a form of the reciprocal-modulus transformation [4, 19.7.3]; only the real part will be needed.

Lemma 2.3 (Bounds)**.** *There is an absolute constant* C *such that, for* $m \in \mathbb{C}\setminus[1,\infty)$ *and also for the boundary values* $\mathbf{K}(m\pm i0)$ *with* $m > 1$*,*

$$|\mathbf{K}(m)| \le \begin{cases} C\big(1+\big|\log|1-m|\big|\big), & |m|\le 4,\\ C\,(1+\log|m|)\,|m|^{-1/2}, & |m|\ge 4.\end{cases}$$

Proof. For $|1-m| \le \frac{1}{2}$ the first bound follows from (2.2), since $\mathbf{K}(1-m)$ and $E(1-m)$ are bounded there and $|\log\frac{1}{1-m}| \le |\log|1-m|| + \pi$. On the remaining part of $\{|m|\le 4\}$, $\mathbf{K}$ is continuous up to both sides of the cut, hence bounded. At ∞ the exponents of (2.1) are $\frac{1}{2}, \frac{1}{2}$, and the nearest other singular point is 1. Thus on $\{|m|>2\}\setminus[2,\infty)$ every solution of (2.1) is a linear combination of $(-m)^{-1/2}\Phi_1(1/m)$ and $(-m)^{-1/2}\big(\Phi_1(1/m)\log(-m) + \Phi_2(1/m)\big)$, where Φ_1, Φ_2 are holomorphic in $|\zeta| < 1$ and the powers and

the logarithm are principal. Applied to $\mathbf{K}$, this gives the second bound, including the boundary values on both sides of $[2,\infty)$. □

We next continue $m \mapsto \mathbf{K}(m)$ along $m = 1 - z^2/16$. Let $\Pi_- = \{z : \operatorname{Im} z < 0\}$ and $\Pi_+ = \{z : \operatorname{Im} z > 0\}$. For $\operatorname{Re} z > 0$ and $\operatorname{Im} z < 0$ we have $z^2 \notin (-\infty, 0]$, hence $1 - z^2/16 \notin [1,\infty)$.

Lemma 2.4 (The continued kernel)**.** *There is a unique holomorphic function P on Π_- such that $P(z) = \frac{1}{2\pi^2}\mathbf{K}(1 - z^2/16)$ when $\operatorname{Re} z > 0$. It has the following properties.*

(a) P extends continuously to $\overline{\Pi_-} \setminus \{0, -4\}$, and for $c > 0$ its boundary value is $P(c) = \frac{1}{2\pi^2}\mathbf{K}(1 - c^2/16) > 0$.

(b) For $0 < c < 4$,

$$P(-c - i0) = f(c) + \frac{i}{\pi^2}\,\mathbf{K}\Big(\frac{c^2}{16}\Big). \tag{2.3}$$

(c) There is a constant C such that for $z \in \overline{\Pi_-} \setminus \{0, -4\}$

$$|P(z)| \le \begin{cases} C\big(1 + \big|\log|z|\big| + \big|\log|z+4|\big|\big), & |z| \le 8, \\ C\,(1 + \log|z|)\,|z|^{-1}, & |z| \ge 8. \end{cases}$$

Proof. Substituting $m = 1 - z^2/16$ into (2.1) gives a linear differential equation of second order in z whose coefficients are holomorphic outside $\{0, 4, -4\}$. The function $z \mapsto \mathbf{K}(1 - z^2/16)$ solves it on $\{\operatorname{Re} z > 0,\ \operatorname{Im} z < 0\}$. Since Π_- is simply connected and contains no singular point, this solution continues uniquely to a holomorphic function on Π_-, and it continues analytically across every real point other than 0 and ± 4. Near $z = 4$ the continuation is the principal function $\mathbf{K}(1 - z^2/16)$, which is holomorphic at $m = 0$. This gives the continuity in (a). For $c > 0$ we have $1 - c^2/16 < 1$, where $\mathbf{K}$ is holomorphic and positive.

For (b), let $z \in \Pi_-$, $|z| < 4$ and $\operatorname{Re} z > 0$. Then $1 - m = z^2/16$, and $\log\frac{1}{1-m} = 2\log\frac{4}{z}$ with principal logarithms, because $\arg z \in (-\pi/2, 0)$. Lemma 2.1 gives

$$2\pi^2 P(z) = \frac{2}{\pi}\,\mathbf{K}\Big(\frac{z^2}{16}\Big)\log\frac{4}{z} + E\Big(\frac{z^2}{16}\Big). \tag{2.4}$$

With the principal branch of $\log z$, the right-hand side is holomorphic on the whole half-disc $\Pi_- \cap \{|z| < 4\}$, so (2.4) holds there. As $z \to c \in (0, 4)$, (2.4) reads $\mathbf{K}(1 - c^2/16) = \frac{2}{\pi}\mathbf{K}(c^2/16)\log\frac{4}{c} + E(c^2/16)$. As $z \to -c$ from Π_-, we have $\log z \to \log c - i\pi$, so $\log\frac{4}{z} \to \log\frac{4}{c} + i\pi$, while $\mathbf{K}(z^2/16) \to \mathbf{K}(c^2/16)$ and $E(z^2/16) \to E(c^2/16)$. Subtracting the two limits gives $2\pi^2 P(-c - i0) = \mathbf{K}(1 - c^2/16) + 2i\,\mathbf{K}(c^2/16)$, which is (2.3).

For (c), (2.4) gives the bound near $z = 0$. Near $z = -4$ we have $m \to 0$, a regular singular point with exponents $\{0, 0\}$, so every solution is $O(1 + |\log|m||)$ there, and $|m| \asymp |z + 4|$. Elsewhere on $\overline{\Pi_-} \cap \{|z| \le 8\}$, P is continuous. For $|z| \ge 8$ we have $|m| \ge 3$. The set $\overline{\Pi_-} \cap \{|z| \ge 8\}$ is simply connected, and on it P is a fixed linear combination of the two solutions at

∞ used in the proof of Lemma 2.3, with $(-m)^{-1/2}$ and $\log(-m)$ continued along this set. Since $\arg(-m) = \arg(z^2/16 - 1)$ varies in a bounded interval there, we obtain $|P(z)| \le C(1 + \log|m|)|m|^{-1/2} \le C'(1 + \log|z|)/|z|$. □

3. The spectral representation

Throughout this section $0 < \alpha \le 1$, $b = 2\sqrt{\alpha}$ and $B = 4 + b$. Let

$$D_n(\alpha) = \alpha^n A_n(1/\alpha) = \sum_{k=0}^{n} \alpha^k \left(\frac{(2n-k)!}{k!\,(n-k)!^2} \right)^2. \tag{3.1}$$

The second expression follows from (1.1) with $k = n - j$, since $\binom{n}{j}\binom{n+j}{j} = \frac{(n+j)!}{j!^2(n-j)!}$. Let $\theta_1, \theta_2, \theta_3$ be independent random variables, uniformly distributed on $[0, 2\pi]$, and let $\mathbb{E}$ denote expectation. Put

$$Y = 4\cos\theta_1 \cos\theta_2, \qquad L = Y + b\cos\theta_3.$$

Proposition 3.1. *For complex τ with $|\tau| < 1/(2B)$,*

$$\sum_{n \ge 0} D_n(\alpha)\, \tau^{2n} = \mathbb{E}\Big[\frac{1}{1 - \tau L + \alpha\tau^2} \Big]. \tag{3.2}$$

Proof. Put $\beta = \sqrt{\alpha}\,\tau$. Then $1 - \tau L + \alpha\tau^2 = (1 - \beta e^{i\theta_3})(1 - \beta e^{-i\theta_3}) - \tau Y$, and

$$\frac{1}{1 - \tau L + \alpha\tau^2} = \sum_{j \ge 0} (\tau Y)^j \sum_{k,l \ge 0} \binom{j+k}{j} \binom{j+l}{j} \beta^{k+l} e^{i(k-l)\theta_3}.$$

Here $|\tau| < 1/8$ and $|\beta| < 1/8$, and $|Y| \le 4$, so the triple series is dominated by $\sum_j (4|\tau|)^j (1 - |\beta|)^{-2j-2} < \infty$, uniformly in the angles. Taking expectations termwise, $\mathbb{E}[e^{i(k-l)\theta_3}] = \delta_{kl}$, and $\mathbb{E}[Y^j] = \mathbb{E}[(2\cos\theta_1)^j]^2$ equals $\binom{2i}{i}^2$ if $j = 2i$ and vanishes for odd j. Hence

$$\mathbb{E}\Big[\frac{1}{1 - \tau L + \alpha\tau^2} \Big] = \sum_{i,k \ge 0} \binom{2i}{i}^2 \binom{2i+k}{k}^2 \alpha^k \tau^{2i+2k}.$$

The coefficient of τ^{2n} is

$$\sum_{k=0}^{n} \alpha^k \binom{2n-2k}{n-k}^2 \binom{2n-k}{k}^2 = \sum_{k=0}^{n} \alpha^k \left(\frac{(2n-k)!}{k!\,(n-k)!^2} \right)^2,$$

which is $D_n(\alpha)$ by (3.1). □

Remark 3.2. Proposition 3.1 is a torus average of Straub's rational function [9, Theorem 3.1]. Substituting $u = \varrho e^{i\theta_1}$, $v = \varrho e^{-i\theta_1}$, $w = \sigma e^{i\theta_2}$, $z = \sigma e^{-i\theta_2}$ in $1/\big((1-u-v)(1-w-z) - \alpha uvwz\big)$ and averaging over θ_1, θ_2 keeps exactly the monomials with equal exponents within each pair, and the diagonal in (ϱ, σ) of the result is again an angular average.

Lemma 3.3. *The random variable Y has the density f of (1.4). In particular, f is even, $f > 0$ on $(-4, 4)$ and $\int f = 1$.*

Proof. The variables $\cos\theta_i$ are independent with the arcsine density $\pi^{-1}(1-y^2)^{-1/2}$ on $(-1,1)$. For $0<a<1$ the density of $\cos\theta_1\cos\theta_2$ at $\pm a$ is

$$2\int_a^1 \frac{1}{r}\cdot\frac{1}{\pi\sqrt{1-r^2}}\cdot\frac{1}{\pi\sqrt{1-a^2/r^2}}\,\mathrm{d}r = \frac{2}{\pi^2}\int_a^1 \frac{\mathrm{d}r}{\sqrt{(1-r^2)(r^2-a^2)}}.$$

The substitution $r^2 = 1-(1-a^2)\sin^2\phi$ gives $\mathrm{d}r/\sqrt{(1-r^2)(r^2-a^2)} = -\,\mathrm{d}\phi/r$, so the last integral equals $\mathbf{K}(1-a^2)$. Since $Y = 4\cos\theta_1\cos\theta_2$, its density at c is $\frac{1}{4}\cdot\frac{2}{\pi^2}\mathbf{K}(1-c^2/16) = f(c)$ for $0<|c|<4$. □

Let $\omega_b(u) = \pi^{-1}(b^2-u^2)^{-1/2}$ for $|u|<b$ and $\omega_b(u)=0$ otherwise; it is the density of $b\cos\theta_3$. Thus L has the density $h = f*\omega_b$, which is (1.5). Let

$$g(w) = \int_{-B}^{B}\frac{h(v)}{w-v}\,\mathrm{d}v \qquad (w\in\mathbb{C}\setminus[-B,B]) \tag{3.3}$$

be its Cauchy transform.

Lemma 3.4. *The density h and its Cauchy transform g have the following properties.*

- *(a) h is even and nonnegative, vanishes outside $(-B,B)$, is positive on $(-B,B)$, and $\int h = 1$.*
- *(b) The function h is Hölder continuous on $\mathbb{R}$ with exponent $1/8$; in particular $h(\pm B)=0$.*
- *(c) g is holomorphic on $\mathbb{C}\setminus[-B,B]$, bounded there, and $|g(w)|\le 2/|w|$ for $|w|\ge 2B$. For real ξ the limits $g(\xi\pm i0) = \lim g(w)$, as $w\to\xi$ within Π_+ or Π_- respectively, exist, depend continuously on ξ, and satisfy*

$$\operatorname{Re} g(\xi+i0) = \operatorname{Re} g(\xi-i0), \qquad \operatorname{Im} g(\xi\pm i0) = \mp\pi h(\xi). \tag{3.4}$$

Proof. (a) All statements except positivity follow from Lemma 3.3. If $|w|<B$, the set of $u\in(-b,b)$ with $|w-u|<4$ is a nonempty open interval, on which $f(w-u)\omega_b(u)>0$.

(b) The function f is bounded by $C(1+|\log|c||)$, satisfies $|f'(c)|\le C(1+|c|^{-1})$ on $(-4,4)\setminus\{0\}$, by Lemma 2.1, and has jumps at ±4. Let $0<|\delta|\le 1$. On the set of points within $2|\delta|$ of $\{0,\pm4\}$, $|f(c+\delta)-f(c)|\le C(1+|\log|c||+|\log|c+\delta||)$, which contributes $O(|\delta|(1+|\log|\delta||)^4)$ to $\|f(\cdot+\delta)-f\|_4^4$. On the complement in $[-5,5]$, the mean value theorem gives $|f(c+\delta)-f(c)|\le C|\delta|(1+|c|^{-1})$, which contributes $O(|\delta|^4\cdot|\delta|^{-3}) = O(|\delta|)$. Thus $\|f(\cdot+\delta)-f\|_4 \le C|\delta|^{1/4}(1+|\log|\delta||)$. Since $\omega_b\in L^{4/3}$, Hölder's inequality yields $|h(w+\delta)-h(w)|\le\|f(\cdot+\delta)-f\|_4\|\omega_b\|_{4/3}\le C'|\delta|^{1/8}$.

(c) Holomorphy and the bound for $|w|\ge 2B$ are clear. Let $w=\xi+iy$ with $0<|y|\le 1$, and split

$$g(w) = \int_{|v-\xi|<1}\frac{h(v)-h(\xi)}{w-v}\,\mathrm{d}v + h(\xi)\int_{-1}^{1}\frac{\mathrm{d}s}{s+iy} + \int_{|v-\xi|\ge 1}\frac{h(v)}{w-v}\,\mathrm{d}v.$$

By (b) the first integrand is bounded by $C|v-\xi|^{-7/8}$ and converges pointwise as $y \to 0$. The middle integral equals $-2i \arctan(1/y)$, which tends to $\mp i\pi$ as $y \to 0^{\pm}$. The last integral is bounded by 1, and $|g(w)| \le 1$ when $|\operatorname{Im} w| \ge 1$. Dominated convergence, which is uniform for ξ in compact sets, gives the boundedness of g, the existence and continuity of the limits, and (3.4), because the limits of the first and the last term are real. □

4. Continuation of the spectral density

From now on $0 < \alpha \le 1$, so that $0 < b \le 2$. For $u \in \mathbb{C} \setminus \big((-\infty, -b] \cup [b, \infty)\big)$ let

$$S_b(u) = \sqrt{b-u}\,\sqrt{b+u}$$

with principal square roots. This is the holomorphic function that equals $\sqrt{b^2-u^2} > 0$ on $(-b, b)$; thus $\omega_b(u) = 1/(\pi S_b(u))$ there. For $w \in \Pi_-$ let $[w, b]$ denote the oriented segment from w to b, and define

$$T_-(w) = \int_{-b}^{b} P(w-u)\,\omega_b(u)\,\mathrm{d}u, \tag{4.1}$$

$$C_-(w) = \frac{1}{\pi} \int_{[w,b]} \mathbf{K}\Big(\frac{(w-u)^2}{16}\Big) \frac{\mathrm{d}u}{S_b(u)}, \tag{4.2}$$

$$H(w) = T_-(w) - \frac{i}{\pi^2}\, C_-(w). \tag{4.3}$$

Lemma 4.1. *The functions T_-, C_- and H are holomorphic on Π_-. For $w \in \Pi_-$,*

$$C_-(w) = \frac{\sqrt{b-w}}{\pi} \int_0^1 \mathbf{K}\Big(\frac{(b-w)^2 v^2}{16}\Big) \frac{\mathrm{d}v}{\sqrt{1-v}\,\sqrt{b+w+(b-w)v}}. \tag{4.4}$$

Proof. For real u we have $w - u \in \Pi_-$, and on compact subsets of Π_- the integrand of (4.1) is holomorphic in w and bounded by a constant times $\omega_b(u)$, by Lemma 2.4(c). Hence T_- is holomorphic.

Parametrize $[w, b]$ by $u = w + (b-w)v$, $0 \le v \le 1$. Then $b - u = (b-w)(1-v)$, and $b - w \in \Pi_+$, so $\sqrt{b-u} = \sqrt{b-w}\sqrt{1-v}$. Moreover $\operatorname{Im}(b+u) = (1-v)\operatorname{Im} w < 0$ for $v < 1$, while $b + u = 2b$ for $v = 1$; hence the path avoids the cuts of S_b. Since $w - u = -(b-w)v$ is not real for $0 < v \le 1$, its square does not lie in $[0, \infty)$, so the argument of $\mathbf{K}$ stays in $\mathbb{C} \setminus [1, \infty)$. This gives (4.4). On compact subsets of Π_- the integrand of (4.4) is holomorphic in w and bounded by a constant times $(1-v)^{-1/2}$, so C_- is holomorphic. □

Proposition 4.2 (The central interval)**.** *H extends continuously to $\Pi_- \cup (-b, b)$, with $H(\xi) = h(\xi)$ for $-b < \xi < b$. Consequently h is real-analytic on $(-b, b)$, H extends to a holomorphic function on $\Pi_- \cup (-b, b) \cup \Pi_+$, again denoted H, and*

$$H(w) = \overline{H(-\overline{w})} \qquad (w \in \Pi_-). \tag{4.5}$$

In particular $H(-iy)$ is real for every $y > 0$.

Proof. Fix $\xi \in (-b, b)$. For $u \in (-b, b)$ the difference $\xi - u$ lies in $(\xi - b, \xi + b)$, and since $b \le 2$,

$$-4 \le -2b < \xi - b < \xi + b < 2b \le 4. \tag{4.6}$$

Let $w \to \xi$ within Π_-. By Lemma 2.4(a),(b), for $u \ne \xi$,

$$P(w-u) \longrightarrow \begin{cases} f(\xi - u), & u < \xi, \\ f(u-\xi) + \dfrac{i}{\pi^2}\mathbf{K}\Big(\dfrac{(u-\xi)^2}{16}\Big), & u > \xi, \end{cases}$$

where we used that f is even and that $u - \xi \in (0, 4)$ for $u > \xi$, by (4.6). By Lemma 2.4(c), and because $w - u$ stays away from -4, we have $|P(w-u)| \le C(1 + |\log|\operatorname{Re} w - u||)$. These bounds, multiplied by ω_b, are uniformly integrable as $\operatorname{Re} w$ varies near ξ: the logarithms are bounded in L^4 and $\omega_b \in L^{4/3}$. Vitali's theorem gives

$$T_-(w) \longrightarrow h(\xi) + \frac{i}{\pi^2}\int_\xi^b \mathbf{K}\Big(\frac{(u-\xi)^2}{16}\Big)\omega_b(u)\,\mathrm{d}u.$$

In (4.4), as $w \to \xi$, we have $b - w \to b - \xi \in (0, 4)$ and $b + w + (b - w)v \to b + \xi + (b - \xi)v > 0$, and the arguments of $\mathbf{K}$ converge to $(b-\xi)^2 v^2/16 \in [0, 1)$, uniformly in v. By dominated convergence and the substitution $u = \xi + (b - \xi)v$,

$$C_-(w) \longrightarrow \int_\xi^b \mathbf{K}\Big(\frac{(u-\xi)^2}{16}\Big)\omega_b(u)\,\mathrm{d}u.$$

Hence $H(w) \to h(\xi)$. The convergence is locally uniform in ξ, and h is continuous, so H is continuous on $\Pi_- \cup (-b, b)$ and real on $(-b, b)$. The Schwarz reflection principle provides the holomorphic extension, and h is real-analytic on $(-b, b)$ as the restriction of H.

The function $w \mapsto \overline{H(-\overline{w})}$ is holomorphic on Π_- and has the boundary values $h(-\xi) = h(\xi)$ on $(-b, b)$, by Lemma 3.4(a). Its difference with H is holomorphic on Π_- with boundary values 0 on $(-b, b)$. By reflection, it extends holomorphically across $(-b, b)$ and vanishes there, hence vanishes identically. This proves (4.5); for $w = -iy$ we have $-\overline{w} = w$. □

Remark 4.3. The inequality $b \le 2$, that is $\alpha \le 1$ or $x \ge 1$, is essential for Proposition 4.2. By (4.6), every difference $\xi - u$ with $\xi, u \in (-b, b)$ lies in $(-4, 4)$, so that $f(\xi - u)$ is given by the single analytic expression (1.4), and the continuation P of Lemma 2.4 applies throughout. If $b > 2$, the differences $\xi - u$ with ξ, u in the central interval can cross ± 4, where f jumps to 0, and the argument breaks down. The same assumption is also used in Propositions 4.4 and 4.5, in the identity $T(b) = h(b)$ and in the lower bounds $|w - u + 4| \ge 2$ and $\operatorname{Re}(1 + dv/4) \ge \frac{1}{2}$, and, for convenience, in Proposition 3.1. We make no assertion for $x < 1$.

To the right of b we use boundary values from below. Let

$$Q = \{w \in \mathbb{C} : \ \operatorname{Re} w \ge 0, \ \operatorname{Im} w \le 0\}.$$

For $\xi > b$ we write $H(\xi - i0) = \lim H(w)$ as $w \to \xi$ within Π_-, and for $m \geq 0$ we write $\mathbf{K}(m - i0)$ for the boundary value from below when $m > 1$ and for $\mathbf{K}(m)$ when $m < 1$.

Proposition 4.4 (Beyond the central interval)**.** *H extends continuously to Q. For $\xi \geq b$,*

$$H(\xi - i0) = T(\xi) + \frac{1}{\pi^3} \int_b^\xi \mathbf{K}\Big(\frac{(\xi-u)^2}{16} - i0\Big) \frac{\mathrm{d}u}{\sqrt{u^2 - b^2}}, \tag{4.7}$$

with T as in (1.6). Consequently $\operatorname{Re} H(\xi - i0) = T(\xi) + J(\xi)$ *with J as in (1.7), and for $\xi > b$*

$$\begin{gathered} 0 \leq h(\xi) \leq T(\xi), \\ 2 \operatorname{Re} H(\xi - i0) - h(\xi) = 2T(\xi) - h(\xi) + 2J(\xi) \geq T(\xi) > 0. \end{gathered} \tag{4.8}$$

Moreover $H(b - i0) = T(b) = h(b)$.

Proof. On $Q \cap \{\operatorname{Re} w < b\}$ near the real axis, continuity follows from Proposition 4.2. Let $\xi \geq b$ and $w \to \xi$ within Π_-. For $u \in (-b, b)$ we have $\xi - u > 0$, so by Lemma 2.4(a) $P(w-u) \to \frac{1}{2\pi^2}\mathbf{K}(1 - (\xi-u)^2/16)$. The same uniform integrability as before gives $T_-(w) \to T(\xi)$.

Consider first the endpoint $\xi = b$. For $u \in (-b, b)$ we have $0 < b - u < 2b \leq 4$, so $f(b-u) = \frac{1}{2\pi^2}\mathbf{K}(1 - (b-u)^2/16)$ and hence $T(b) = h(b)$. Let $w \in \Pi_-$ with $|w - b| \leq \min(b/2, 2)$. In (4.4) the arguments of $\mathbf{K}$ satisfy $|(b-w)^2 v^2/16| \leq \frac{1}{4}$, where $\mathbf{K}$ is bounded, and $|b + w + (b-w)v - 2b| = |w-b|(1-v) \leq b/2$, so that $|b+w+(b-w)v| \geq b$. Since $\int_0^1 (1-v)^{-1/2}\,\mathrm{d}v = 2$, we obtain $|C_-(w)| \leq C|b-w|^{1/2}$, and the same bound holds for the boundary values on $(b, b + \min(b/2, 2)]$. Hence $C_-(w) \to 0$ and $H(w) \to T(b) = h(b)$ as $w \to b$ within Q. No sign of $\operatorname{Im}(b-w)^2$ is used here; indeed this sign is not constant near b, since $w = b - \varepsilon - i\varepsilon$ gives $(b-w)^2 = 2i\varepsilon^2$.

Now let $\xi > b$. In (4.4), $b - w$ tends to the negative number $b - \xi$ from Π_+, so $\sqrt{b-w} \to i\sqrt{\xi - b}$. Also $b + w + (b-w)v \to 2b + (\xi - b)(1-v) > 0$. Finally $\operatorname{Im}(b-w)^2 = 2\operatorname{Re}(b-w)\operatorname{Im}(b-w) < 0$ near ξ, so the arguments of $\mathbf{K}$ approach $(\xi-b)^2 v^2/16$ from the lower half-plane. By Lemma 2.3 the integrands are bounded by a constant times $(1 + |\log|1 - \lambda v||)(1-v)^{-1/2}$ with $\lambda = |b-w|/4$, and this family is uniformly integrable for λ in compact sets. Hence

$$C_-(w) \longrightarrow \frac{i\sqrt{\xi - b}}{\pi} \int_0^1 \mathbf{K}\Big(\frac{(\xi-b)^2 v^2}{16} - i0\Big) \frac{\mathrm{d}v}{\sqrt{1-v}\sqrt{2b + (\xi-b)(1-v)}},$$

and the substitution $u = b + (\xi - b)(1-v)$ turns the right-hand side into

$$\frac{i}{\pi} \int_b^\xi \mathbf{K}\Big(\frac{(\xi-u)^2}{16} - i0\Big) \frac{\mathrm{d}u}{\sqrt{u^2 - b^2}}.$$

Multiplying by $-i/\pi^2$ gives (4.7). The convergence is locally uniform in $\xi > b$; together with the endpoint case and Proposition 4.2, this proves continuity on Q.

By Lemma 2.2, $\operatorname{Re}\mathbf{K}(m-i0)=\kappa(m)$ for $m\ge 0$, $m\neq 1$; hence $\operatorname{Re}H(\xi-i0)=T(\xi)+J(\xi)$, and $J\ge 0$ because $\kappa>0$. For $\xi>b$ and $u\in(-b,b)$ we have $\xi-u>0$, and $f(\xi-u)=\frac{1}{2\pi^2}\mathbf{K}(1-(\xi-u)^2/16)$ if $\xi-u<4$, while $f(\xi-u)=0$ otherwise. Since $\mathbf{K}$ is positive on $(-\infty,1)$, this gives $0\le h(\xi)\le T(\xi)$, and (4.8) follows, with $T(\xi)>0$. For $\xi=b$ the integral in (4.7) is empty, and (4.7) reduces to $H(b-i0)=T(b)=h(b)$, proved above. □

Proposition 4.5 (Bounds)**.** *For every $R>0$, H is bounded on $Q\cap\{|w|\le R\}$. There are constants C and R_0, depending on α, such that*

$$|H(w)|\le C\,\frac{\log^2|w|}{|w|}\qquad(w\in Q,\ |w|\ge R_0). \tag{4.9}$$

Proof. By continuity it suffices to consider $w\in Q\cap\Pi_-$. For $u\in(-b,b)$ we have $\operatorname{Re}(w-u)>-b\ge -2$, hence $|w-u+4|\ge 2$. If $|w|\le R$, Lemma 2.4(c) gives $|P(w-u)|\le C_R(1+|\log|\operatorname{Re}w-u||)$, and $\sup_{t\in\mathbb{R}}\int(1+|\log|t-u||)\omega_b(u)\,\mathrm{d}u<\infty$; so T_- is bounded. If $|w|\ge 16$, then $|w-u|\ge|w|/2\ge 8$ and $|P(w-u)|\le C\log|w|/|w|$, so $T_-(w)=O(\log|w|/|w|)$.

For C_- write $d=w-b$ and $D=|d|$, so that $\operatorname{Re}d\ge -b$. For $0\le v\le 1$ the number $\zeta=d(1-v)$ satisfies $\operatorname{Re}(2b+\zeta)\ge b$ and $|2b+\zeta|\ge|\zeta|-2b$; distinguishing $|\zeta|\le 3b$ and $|\zeta|>3b$ one obtains $|2b+\zeta|\ge c_b(1+|\zeta|)$ with $c_b=b/(1+3b)$. Moreover the argument $m=d^2v^2/16$ of $\mathbf{K}$ in (4.4) satisfies $1-m=(1-dv/4)(1+dv/4)$ with $\operatorname{Re}(1+dv/4)\ge 1-b/4\ge\frac12$ and $|1-dv/4|\ge|1-Dv/4|$. Therefore (4.4) gives

$$|C_-(w)|\le\frac{1}{\pi\sqrt{c_b}}\int_0^1|\mathbf{K}(m)|\,\frac{\sqrt{D}\,\mathrm{d}v}{\sqrt{1-v}\,\sqrt{1+D(1-v)}}. \tag{4.10}$$

If $D\le R$, Lemma 2.3 and the lower bound for $|1-m|$ give $|\mathbf{K}(m)|\le C_R(1+|\log|1-Dv/4||)$, and $\sup_{0\le\lambda\le R}\int_0^1(1+|\log|1-\lambda v||)(1-v)^{-1/2}\,\mathrm{d}v<\infty$ by Hölder's inequality. Hence C_- is bounded on bounded parts of Q.

Let $D\ge 16$ and split $[0,1]$ into $I_1=[0,8/D]$, $I_2=[8/D,1-1/D]$ and $I_3=[1-1/D,1]$.

- On I_1 we have $1-v\ge\frac12$, so the weight in (4.10) is at most 2, and $|m|\le 4$, so $|\mathbf{K}(m)|\le C(1+|\log|1-Dv/4||)$. The contribution is at most $C\int_0^{8/D}(1+|\log|1-Dv/4||)\,\mathrm{d}v=O(1/D)$. A possible crossing of $Dv=4$ contributes only an integrable logarithm.
- On I_2 we have $|m|^{1/2}=Dv/4\ge 2$, so $|\mathbf{K}(m)|\le C\log D/(Dv)$, and the weight is at most $(1-v)^{-1}$. The contribution is at most $C\frac{\log D}{D}\int_{8/D}^{1-1/D}\frac{\mathrm{d}v}{v(1-v)}=O\big(\frac{\log^2 D}{D}\big)$.
- On I_3 we have $|m|\ge D^2/64\ge 4$, so $|\mathbf{K}(m)|\le C\log D/D$, and the weight is at most $\sqrt{D}(1-v)^{-1/2}$, whose integral over I_3 equals 2. The contribution is $O(\log D/D)$.

Since $D\ge|w|-b$, we obtain $C_-(w)=O(\log^2|w|/|w|)$, and (4.9) follows from (4.3). □

5. The Stieltjes transform and the proof of Theorem 1.1

Let $0 < \alpha \le 1$, and let $S > \alpha$ be determined by $\sqrt{S} + \alpha/\sqrt{S} = B$, that is,

$$S = \left(\frac{B + \sqrt{B^2 - 4\alpha}}{2} \right)^2. \tag{5.1}$$

For $s \in \mathbb{C} \setminus (-\infty, 0]$ let $q = q(s) = \sqrt{s}$ be the principal square root, and put

$$r = r(s) = q + \frac{\alpha}{q}.$$

Lemma 5.1. *The map $s \mapsto r(s)$ has the following properties.*

(a) Let $s \in \Pi_+$. Then $\operatorname{Re} r > 0$ and $\operatorname{sign} \operatorname{Im} r = \operatorname{sign}(|s| - \alpha)$. If $|s| = \alpha$, then $r \in (0, b)$. If $|s| < \alpha$, then r lies in the interior of Q.

(b) r maps $(0, \alpha]$ decreasingly onto $[b, \infty)$, $[\alpha, S]$ increasingly onto $[b, B]$, and (S, ∞) onto (B, ∞).

(c) As $s \to -a$ from Π_+, with $a > 0$, we have $q \to i\sqrt{a}$ and $r \to i(\sqrt{a} - \alpha/\sqrt{a})$.

(d) If $s \in \Pi_+$ and $|s| \le \alpha/4$, then $|r| \ge 3\alpha/(4|q|)$ and $|r|\,|q| \ge 3\alpha/4$.

Proof. For $s \in \Pi_+$ the root q lies in the open first quadrant, and $\operatorname{Re} r = \operatorname{Re} q\,(1 + \alpha/|q|^2) > 0$, $\operatorname{Im} r = \operatorname{Im} q\,(1 - \alpha/|q|^2)$. If $|q| = \sqrt{\alpha}$, then $q = \sqrt{\alpha} e^{i\phi}$ with $0 < \phi < \pi/2$ and $r = 2\sqrt{\alpha}\cos\phi \in (0, b)$. Parts (b) and (c) are elementary. For (d), $|q|^2 \le \alpha/4$ gives $|q| \le \alpha/(4|q|)$, so $|r| \ge \alpha/|q| - |q| \ge 3\alpha/(4|q|)$; and $|r||q| = |s + \alpha| \ge 3\alpha/4$. □

Define $\widehat{G}$ on $\Pi_+ \cup \Pi_-$ by

$$\widehat{G}(w) = g(w) \quad (w \in \Pi_+), \qquad \widehat{G}(w) = g(w) - 2\pi i\, H(w) \quad (w \in \Pi_-).$$

Lemma 5.2. *$\widehat{G}$ extends holomorphically across $(-b, b)$, and continuously to the point b from both Π_+ and Q.*

Proof. For $\xi \in (-b, b)$, Lemma 3.4(c) and Proposition 4.2 give the limits $g(\xi + i0)$ from above and $g(\xi - i0) - 2\pi i h(\xi)$ from below. By (3.4) these coincide, and they depend continuously on ξ. Hence $\widehat{G}$ is continuous across $(-b, b)$, and holomorphic there by Morera's theorem. At b, the limit from Π_+ is $g(b+i0)$ and the limit from Q is $g(b-i0) - 2\pi i H(b-i0) = g(b-i0) - 2\pi i h(b)$, by Proposition 4.4; again these coincide. □

For $s \in \Pi_+$ put

$$M(s) = \frac{\widehat{G}(r(s))}{q(s)} = \begin{cases} g(r)/q, & |s| > \alpha, \\ \big(g(r) - 2\pi i H(r)\big)/q, & |s| < \alpha. \end{cases} \tag{5.2}$$

By Lemmas 5.1(a) and 5.2, M is holomorphic on Π_+; across the semicircle $|s| = \alpha$ the two lines of (5.2) are continuations of each other.

Lemma 5.3. *The series $M_\infty(s) = \sum_{n \ge 0} D_n(\alpha) s^{-n-1}$ converges for $|s| > 4B^2$, and $M = M_\infty$ on $\Pi_+ \cap \{|s| > 4B^2\}$.*

Proof. Let $|s| > 4B^2$ and $\tau = 1/q$, so that $|\tau| < 1/(2B)$; then Proposition 3.1 gives the convergence. For $v \in [-B, B]$,

$$\frac{1}{1 - \tau v + \alpha\tau^2} = \frac{q^2}{q^2 - vq + \alpha} = \frac{q}{r - v}.$$

Integrating against h and using (3.2), we get $\sum_n D_n(\alpha) s^{-n} = q\, g(r)$. Dividing by $s = q^2$, and noting $|s| > \alpha$, gives the claim. □

Define ρ_α on $(0, S)$ by

$$\rho_\alpha(t) = \begin{cases} \dfrac{h(r(t))}{\sqrt{t}}, & \alpha \le t < S, \\ \dfrac{2T(r(t)) - h(r(t)) + 2J(r(t))}{\sqrt{t}}, & 0 < t < \alpha. \end{cases} \tag{5.3}$$

Proposition 5.4. *M extends to a holomorphic function on $\mathbb{C} \setminus [0, S]$ with $M(\overline{s}) = \overline{M(s)}$ and $M = M_\infty$ for $|s| > 4B^2$. Moreover:*

(a) every $s_0 \in (0, S]$ has a neighbourhood N such that M is bounded on $N \setminus [0, S]$, and there is C with $|M(s)| \le C(1 + \log^2 |s|)$ for $0 < |s| \le \alpha/4$, $s \notin [0, S]$;

(b) $M(s) = 1/s + O(|s|^{-2})$ as $s \to \infty$;

(c) for $t \in (0, S)$ the limits $M(t \pm i0)$ exist, are continuous in t, satisfy $M(t - i0) = \overline{M(t + i0)}$, and $-\frac{1}{\pi} \operatorname{Im} M(t + i0) = \rho_\alpha(t)$.

Proof. We first show that M extends continuously from Π_+ to $(-\infty, 0) \cup (S, \infty)$ with real values. For $t > S$, $r(t) \in (B, \infty)$ by Lemma 5.1(b), where g is holomorphic and real; so $M(t) = g(r(t))/\sqrt{t}$ is real. Let $s \to -a$ with $a > 0$, and put $y = \sqrt{a} - \alpha/\sqrt{a}$. By Lemma 5.1(c), $M(s) \to \widehat{G}(iy)/(i\sqrt{a})$, where $\widehat{G}$ is continuous at iy (for $y = 0$ by Lemma 5.2). If $y \ne 0$, the evenness of h gives

$$g(iy) = \int \frac{h(v)(-iy - v)}{y^2 + v^2}\, \mathrm{d}v = -iy \int \frac{h(v)}{y^2 + v^2}\, \mathrm{d}v \in i\mathbb{R}.$$

If $y > 0$, then $\widehat{G}(iy) = g(iy) \in i\mathbb{R}$. If $y < 0$, then $\widehat{G}(iy) = g(iy) - 2\pi i H(iy) \in i\mathbb{R}$, because $H(iy)$ is real by Proposition 4.2. If $y = 0$, $\widehat{G}(0) \in i\mathbb{R}$ by continuity. In all cases $M(-a)$ is real. By the Schwarz reflection principle, M extends holomorphically to $\mathbb{C} \setminus [0, S]$ with $M(\overline{s}) = \overline{M(s)}$. Lemma 5.3, the reality of the coefficients $D_n(\alpha)$ and continuity give $M = M_\infty$ for $|s| > 4B^2$, and (b) follows because $D_0(\alpha) = 1$.

(a) Let $s_0 \in (0, S]$ and let $s \in \Pi_+$ be close to s_0. Then $|q|$ is bounded below. If $|s| \ge \alpha$, then $|\widehat{G}(r)|$ is bounded by Lemmas 3.4(c) and 5.2. If $|s| < \alpha$, then $r \in Q$ is bounded, and $|\widehat{G}(r)| \le |g(r)| + 2\pi |H(r)|$ is bounded by Lemma 3.4(c) and Proposition 4.5. By reflection the same holds in Π_-. Near 0, let $s \in \Pi_+$ with $|s| \le \alpha/4$ and so small that $|r| \ge \max(2B, R_0)$, which is possible by Lemma 5.1(d). Then $|g(r)| \le 2/|r|$ and $|H(r)| \le C \log^2 |r|/|r|$.

By Lemma 5.1(d), $|r||q| \geq 3\alpha/4$, and $|r| \leq |q| + \alpha/|q| \leq 2\alpha/|q|$, so $\log|r| \leq C(1 + |\log|s||)$. Hence

$$|M(s)| \leq \frac{2 + 2\pi C \log^2|r|}{|r|\,|q|} \leq C'(1 + \log^2|s|).$$

On the remaining compact part of $\{|s| \leq \alpha/4\}$ the bound follows from the first assertion of (a), and the case $\operatorname{Im} s < 0$ follows by reflection.

(c) Let $t \in (0, S)$ and $s \to t$ from Π_+; then $q \to \sqrt{t}$. If $t > \alpha$, then by Lemma 5.1(a),(b) the point $r(s)$ tends to $r(t) \in (b, B)$ from Π_+, so $M(t + i0) = g(r(t) + i0)/\sqrt{t}$, and (3.4) gives $-\frac{1}{\pi}\operatorname{Im} M(t + i0) = h(r(t))/\sqrt{t}$. If $t < \alpha$, then $r(s)$ tends to $r(t) \in (b, \infty)$ within Q. Since H is continuous on Q (Proposition 4.4),

$$M(t + i0) = \frac{g(r(t) - i0) - 2\pi i\, H(r(t) - i0)}{\sqrt{t}}.$$

By (3.4), $\operatorname{Im} g(r - i0) = \pi h(r)$, and $\operatorname{Im}(-2\pi i H) = -2\pi \operatorname{Re} H$. Therefore

$$-\frac{1}{\pi}\operatorname{Im} M(t + i0) = \frac{2\operatorname{Re} H(r(t) - i0) - h(r(t))}{\sqrt{t}},$$

which equals the second line of (5.3) by Proposition 4.4. If $t = \alpha$, then $r(s) \to b$, and Lemma 5.2 gives $M(\alpha + i0) = g(b + i0)/\sqrt{\alpha}$. Both lines of (5.3) tend to $h(b)/\sqrt{\alpha}$ as $t \to \alpha$, since $T(b) = h(b)$ and $J(b) = 0$. Continuity in t follows from the continuity of $g(\cdot \pm i0)$, of H on Q, and of $\widehat{G}$. The relation $M(t - i0) = \overline{M(t + i0)}$ is a consequence of $M(\overline{s}) = \overline{M(s)}$. □

Theorem 5.5. *Let $0 < \alpha \leq 1$. The function ρ_α of (5.3) is strictly positive on $(0, S)$, satisfies $\rho_\alpha(t) = O(1 + \log^2 t)$ as $t \to 0^+$, and is integrable on $(0, S)$. Moreover*

$$M(s) = \int_0^S \frac{\rho_\alpha(t)}{s - t}\,\mathrm{d}t \qquad (s \in \mathbb{C} \setminus [0, S]). \tag{5.4}$$

Consequently $\int_0^S \rho_\alpha(t)\,\mathrm{d}t = 1$ and

$$D_n(\alpha) = \int_0^S t^n \rho_\alpha(t)\,\mathrm{d}t \qquad \text{for all } n \geq 0. \tag{5.5}$$

Proof. For $\alpha \leq t < S$ we have $r(t) \in [b, B)$, where $h > 0$ by Lemma 3.4(a). For $0 < t < \alpha$, (4.8) gives $\rho_\alpha(t) \geq T(r(t))/\sqrt{t} > 0$. Since $\rho_\alpha(t) = -\frac{1}{\pi}\operatorname{Im} M(t + i0)$, Proposition 5.4(a) gives $|\rho_\alpha(t)| \leq C(1 + \log^2 t)$ near 0, and local boundedness on $[\varepsilon, S)$; hence ρ_α is integrable.

Fix $s_0 \in \mathbb{C} \setminus [0, S]$. For $0 < \varepsilon < \min(\alpha/4, 1)$ let $U_\varepsilon = \{z : \operatorname{dist}(z, [0, S]) < \varepsilon\}$, and let $R > |s_0| + S + 1$. Cauchy's integral formula on $\{|z| < R\} \setminus \overline{U_\varepsilon}$, for ε small, gives

$$M(s_0) = \frac{1}{2\pi i}\oint_{|z|=R} \frac{M(z)}{z - s_0}\,\mathrm{d}z - \frac{1}{2\pi i}\oint_{\partial U_\varepsilon} \frac{M(z)}{z - s_0}\,\mathrm{d}z,$$

both contours being oriented counterclockwise. The first integral is $O(1/R)$ by Proposition 5.4(b) and tends to 0 as $R \to \infty$. The boundary ∂U_ε consists of the segments $[0, S] \pm i\varepsilon$ and of two semicircles of radius ε centred at 0 and S. By Proposition 5.4(a), the integral over the semicircle at S is $O(\varepsilon)$, and the integral over the semicircle at 0 is $O(\varepsilon \log^2 \varepsilon)$; both tend to 0. On the segments, the integrands converge pointwise for $t \in (0, S)$ to $M(t \pm i0)/(t - s_0)$, by Proposition 5.4(c), and they are dominated by $C(1 + \log^2 t)$, by Proposition 5.4(a). In the counterclockwise orientation the lower segment is traversed from 0 to S and the upper one from S to 0. By dominated convergence,

$$-\frac{1}{2\pi i} \oint_{\partial U_\varepsilon} \frac{M(z)}{z - s_0}\, \mathrm{d}z \longrightarrow -\frac{1}{2\pi i} \int_0^S \frac{M(t - i0) - M(t + i0)}{t - s_0}\, \mathrm{d}t.$$

By Proposition 5.4(c), $M(t - i0) - M(t + i0) = -2i \operatorname{Im} M(t + i0) = 2\pi i \rho_\alpha(t)$, and the right-hand side equals $\int_0^S \rho_\alpha(t)(s_0 - t)^{-1}\, \mathrm{d}t$. This proves (5.4). In particular, no atom, pole or additional holomorphic term is present.

Multiplying (5.4) by s and letting $s \to +\infty$, dominated convergence and Proposition 5.4(b) give $\int_0^S \rho_\alpha = 1$. For $|s| > S$ we may expand $(s - t)^{-1} = \sum_{n \geq 0} t^n s^{-n-1}$ uniformly in $t \in [0, S]$, so $M(s) = \sum_{n \geq 0} s^{-n-1} \int_0^S t^n \rho_\alpha(t)\, \mathrm{d}t$. Comparing with $M = M_\infty$ for $|s| > 4B^2$ and using the uniqueness of Laurent coefficients gives (5.5). □

Proof of Theorem 1.1. Let $x \geq 1$ and $\alpha = 1/x \in (0, 1]$. Then $b = 2\sqrt{\alpha} = 2x^{-1/2}$, in accordance with the definitions of the introduction. By (5.1),

$$\sqrt{S/\alpha} = \frac{B}{2\sqrt{\alpha}} + \sqrt{\frac{B^2}{4\alpha} - 1} = (1 + 2\sqrt{x}) + \sqrt{(1 + 2\sqrt{x})^2 - 1}$$

and the right-hand side equals $1 + 2\sqrt{x} + 2\sqrt{x + \sqrt{x}}$, so $S/\alpha = c_x$. Let μ_x be the image of $\rho_\alpha(t)\, \mathrm{d}t$ under $t \mapsto t/\alpha$. Its density is $\alpha \rho_\alpha(\alpha t)$ on $(0, c_x)$. Since $r(\alpha t) = \sqrt{\alpha t} + \sqrt{\alpha/t} = r_x(t)$ and $\alpha/\sqrt{\alpha t} = 1/\sqrt{xt}$, and since $\alpha t < \alpha$ exactly when $t < 1$, this density is the function φ_x of (1.9). By (3.1) and Theorem 5.5,

$$A_n(x) = x^n D_n(1/x) = \alpha^{-n} \int_0^S t^n \rho_\alpha(t)\, \mathrm{d}t = \int_0^{c_x} t^n \varphi_x(t)\, \mathrm{d}t$$

for all $n \geq 0$. By Theorem 5.5, μ_x is a probability measure, φ_x is strictly positive on $(0, c_x)$, so that the support of μ_x is $[0, c_x]$, and $\varphi_x(t) = O(\log^2 t)$ as $t \to 0^+$.

For uniqueness, let ν be a positive Borel measure on $\mathbb{R}$ with $\int t^n\, \mathrm{d}\nu = A_n(x)$ for all n. For $\lambda > c_x$ we get $\nu(\{|t| \geq \lambda\}) \leq \lambda^{-2n} A_{2n}(x) \leq (c_x/\lambda)^{2n} \to 0$, so ν is supported in $[-c_x, c_x]$. There, polynomials are dense in the continuous functions, and ν and μ_x agree on polynomials; hence $\nu = \mu_x$. □

6. Further remarks

Remark 6.1 (Consequences). For each fixed real $x \geq 1$, the Hankel matrix $(A_{i+j}(x))_{i,j\geq 0}$ is totally positive [10], and, by the theorem of Wang and Zhu that Stieltjes moment sequences are infinitely log-convex [10], the sequence $(A_n(x))_{n\geq 0}$ is infinitely log-convex. The coefficientwise statement of [8, Conjecture 2] does not follow.

Remark 6.2 (Lattice Green functions). Since $4\cos\theta_1\cos\theta_2 = 2\cos(\theta_1+\theta_2)+2\cos(\theta_1-\theta_2)$ and $(\theta_1,\theta_2)\mapsto(\theta_1+\theta_2,\theta_1-\theta_2)$ preserves Haar measure on the torus, for fixed real $x \geq 1$ the variable L has the law of $2(\cos\eta_1+\cos\eta_2+\sqrt{\alpha}\cos\eta_3)$ with independent uniform η_i. Thus $g(w) = \mathbb{E}[(w-L)^{-1}]$ is, up to normalization, the Green function at the origin of the anisotropic simple cubic lattice with anisotropy parameter $\sqrt{\alpha} = x^{-1/2}$, studied by Delves and Joyce [2, 3]. In this language f is the density of states of the square lattice.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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